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CPBIS: Achieving Theoretical Minimum Discovery-latency in Offline Finding Network via Certain Proportion Broadcast Intervals Screening Mechanism

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Abstract

In the Offline Finding Network (OFN), lost items are located using offline Bluetooth Low Energy (BLE) tags attached to the items. A critical component of the OFN is the BLE Neighbor Discovery Process (NDP). However, the high discovery latency of existing methods significantly impacts the user experience of locating lost items. Our research demonstrates that, in most scenarios, the dual-interval broadcast mode achieves lower discovery latency compared to the current single-interval broadcast mode, with only rare cases where the dual-interval mode degenerates into a single-interval one. To address this, we propose a Certain Proportion Broadcast Intervals Screening (CPBIS) mechanism, which achieves theoretical minimum discovery latency for the multiple scan modes. CPBIS consists of three key modules: Mixed Distribution Acquisition, Eliminate Non-Increasing Elements and Calculate Optimal Interval Pair. We have formally validated the optimality of CPBIS through theoretical proof and evaluated its performance on the BLE nRF52832 chip. The experimental results show that CPBIS significantly reduces the discovery latencies and achieves higher discovery success rates under various scan modes, outperforming the existing baseline methods by 31.3% and 7.3% respectively.

Keywords: BLE Neighbor Discovery, Offline BLE Tag, Broadcast Interval Screening, Bluetooth Low Energy

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1. Introduction

The item separation detection has been featured by numerous vendors, including Apple [1], Huawei [2], and Samsung [3], through the Bluetooth Low Energy (BLE) based Offline Finding Network (OFN). Specifically, a BLE tag attached to the item periodically broadcasts neighbor discovery packets, which are captured by the finder device and forwarded to the cloud. The cloud continuously monitors the BLE tag's location based on the discovery data from the finder device and compares it with the owner's location to detect separation. If the owner moves away from the item, a separation alert is generated and sent to the owner's device, such as a smartphone. The effectiveness of item separation detection primarily depends on the BLE neighbor discovery latency, which is influenced by three key factors: the BLE tag's broadcast interval, the scan interval, and the scan window size of the finder devices. Since the BLE tag is energy-constrained compared to the finder device, it is crucial to minimize the tag discovery latency resulting from NDP broadcasts, given the limited battery capacity.

According to previous research [4], reducing the BLE tag broadcast interval results in lower discovery latency, while increasing the energy consumption of the BLE tag. To keep the BLE tag's energy consumption within acceptable limits, the broadcast interval must meet a minimum interval constraint. Thus, the discovery latency optimization problem becomes a minimization problem subject to this constraint. Existing studies [5–8] have optimized the broadcast interval by leveraging the broadcast interval-discovery latency distribution generated by Blender [9], a BLE neighbor discovery simulator. These approaches identify the optimal broadcast interval for the single broadcast pattern.

This paper presents Certain Proportion Broadcast Intervals Screening mechanism (CPBIS), a novel algorithm to select the broadcast interval optimally. Inspired by a preliminary analysis of the broadcast interval-discovery latency distribution, CPBIS employs a dual broadcast pattern to further reduce discovery latency. We have theoretically proven that the neighbor discovery process using the dual broadcast pattern achieves lower discovery latency compared to broadcast patterns employing any other number of intervals.

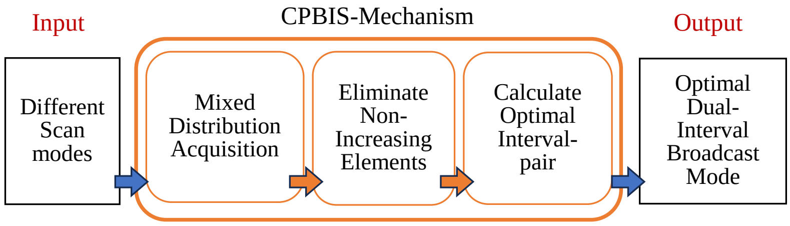


Figure 1: CPBIS takes different BLE scan modes as input, processes them mainly through three internal modules, and finally outputs the optimal dual-interval broadcast mode for offline BLE tag.

CPBIS consists of the following three modules: Mixed Distribution Acquisition, Eliminate Non-Increasing Elements, Calculate Optimal Interval Pair, as shown in

Figure 1. Specifically, **Mixed Distribution Acquisition** superimposes the *Broadcast Interval-Discovery Latency* distributions with different scan modes. Firstly, we use the simulator Blender to generate the *Broadcast Interval–Discovery Latency* distribution for each scan mode. Based on the impact of each scan mode, the Mixed distribution is formed by superimposing the distribution for each scan mode by the percentage of different finder devices. **Eliminate Non-Increasing Elements** removes non-increasing points in the mixed distribution since non-increasing points in the mixed distribution cannot yield the optimal result. Removing those points can significantly reduce the scale of the solution space and the computational complexity. **Calculate Optimal Interval Pair** obtains the optimal dual broadcast intervals and the proportion of each broadcast interval. The optimality of the resulting intervals has been theoretically validated. We have evaluated the CPBIS mechanism through BLE neighbor discovery experiments conducted on the Nordic nRF52832 chip [10]. We test the broadcast mode with two broadcast intervals obtained by CPBIS in environments with different levels of interference and compare it with five other broadcast modes. The results show that the proposed CPBIS mechanism outperforms the baseline methods by 31.3% and 7.3% in terms of the weighted average discovery latency and the neighbor discovery success rate respectively.

In summary, we make the following major contributions:

- We prove that the broadcast mode attaining the theoretical minimum weighted average discovery latency requires no more than two broadcast intervals. In most cases, the weighted average discovery latency generated by an appropriate dual-interval broadcast mode is lower than that of a single-interval broadcast mode.
- We propose CPBIS mechanism, a complete dual-interval broadcast screening mechanism. It solves this problem with three steps: Mixed Distribution Acquisition, Eliminate Non-Increasing Elements and Calculate Optimal Interval Pair.
- We mathematically prove the optimality of the dual-interval broadcast mode derived from the CPBIS mechanism and propose a slope-based strategy along with a Lower Convex Hull method to determine the two optimal broadcast intervals and their corresponding proportions. Furthermore, extensive experiments validate the effectiveness and performance of the CPBIS mechanism.

2. Background

2.1. Offline Finding Network

OFN consists of four main components, which are BLE tags, cloud servers, owner devices and finder devices. Figure 2 illustrates the detailed working mechanism in OFN. Initially, the BLE tag is paired with the owner's device. When the owner's device is within range, the tag remains in a normal state. However, once the tag moves out of the connection range, it switches to a disconnected state. If the tag remains out of range exceed a certain period, it begins to broadcast packets to its surroundings based on a predefined broadcast mode. Once a nearby finder device gets the packet, it will send the location of BLE tag to the cloud server. Furthermore the cloud server push this information to the ower's device. For example, in Apple's AirTag, the tag broadcasts its identity information and public key every 2000 ms. When a nearby finder device within the "Find My" network [11] successfully captures one of the tag's broadcast packets via BLE Neighbor Discovery Protocol (NDP), it uploads encrypted information, including location details, to the cloud server [12]. The cloud server subsequently forwards this information to the owner [13].

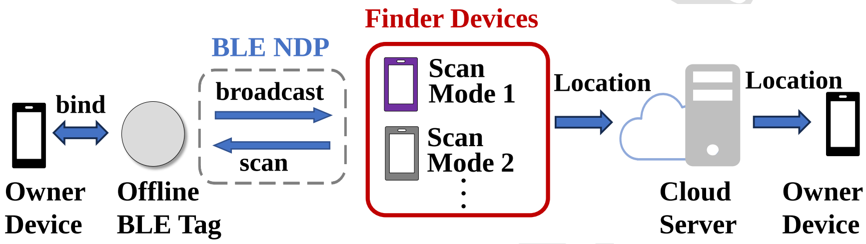


Figure 2: The owner device first binds with the offline BLE tag. After the tag is lost, the Finder Device discovers the tag through BLE NDP and reports its location to the cloud server, which then sends the location to the Owner Device.

To date, numerous studies on Offline Finding Network have been conducted, with most focusing on the privacy and security aspects of OFN [14–18]. Unlike these works, our paper centers on the BLE tag's offline finding functionality. Specifically, our goal is to reduce discovery latency and enhance the success rate in the BLE neighbor discovery process.

2.2. BLE Neighbor Discovery Process

BLE neighbor discovery is a critical technology in OFN. The BLE tag functions as a broadcast device, while the finder devices serve as BLE scan devices. The BLE neighbor discovery process of a finder device involves both broadcasting and scanning. This process begins with the time when the finder device enters the communication range of the BLE tag and ends with the time when it successfully captures a broadcast packet or leaves the range.

For a broadcast device, the key parameter is the broadcast interval A, indicating that the device broadcasts externally at regular intervals of A. However, in practical applications, after a BLE broadcast device determines its broadcast interval, a random advertising delay is added to each broadcast interval. This serves as a random anti-collision protocol, preventing periodic simultaneous transmissions that may occur when multiple devices broadcast in the same environment, which could otherwise cause mutual interference. The main parameters of a scan device are the scan interval T and the scan window W. The scan device performs a scan once every T time units, with W representing the active scan duration within each working

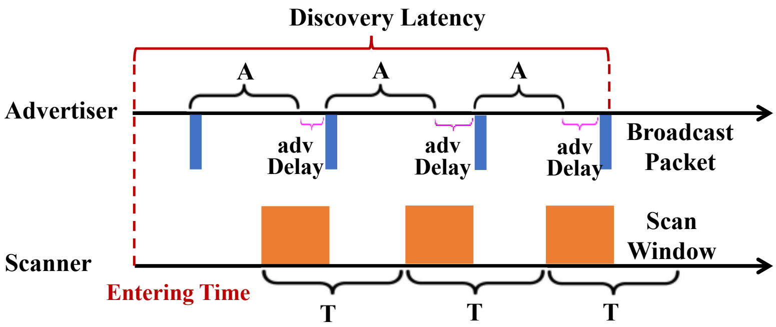


Figure 3: BLE neighbor discovery process, Advertiser refers to the broadcast device, and Scanner refers to the scan device. The discovery latency is defined as the time from the moment the Scanner enters the broadcast range until it captures a complete broadcast packet.

cycle. Therefore, W is configured to be smaller than T. A random BLE neighbor discovery process is shown in Figure 3.

In practical scenarios, the scan device may enter the broadcast range of the broadcast device at random times. Figure 3 illustrates the method for calculating the discovery latency in the BLE NDP of the OFN. Discovery latency is defined as the time elapsed from when a pedestrian carrying the finder device enters the broadcast range of the offline tag until the scan window of finder device successfully receives a complete broadcast packet sent by the tag.

2.3. Broadcast Interval-Discovery Latency Distribution

This section focuses on the research basis of CPBIS, which helps to understand how the CPBIS mechanism works in the next section. Here, we introduce the process of obtaining the "Broadcast Interval - Discovery Latency" distribution. First, in order to optimize the discovery latency of BLE NDP by focusing on broadcast mode, it is necessary to understand how the setting of the broadcast interval affects the value of discovery latency. Using the mathematical model provided in [9], a Cumulative Distribution Function graph of discovery latency will be generated under specific parameters (given the values of T, W, A), as shown in the first step of Figure 4. However, a single CDF graph is insufficient to reveal the relationship between broadcast interval and discovery latency. Therefore, we initialize the broadcast interval A with a starting value (e.g., 50 ms) and define a step size (e.g., 5 ms) to continuously increase A while keeping in a fixed scan mode (T, W). This allows us to generate multiple CDF curves with respect to different A.

Since the value of A approaches infinity as the discovery probability P approaches 100%, we consider values of P in the range of 90% to 95%. Then, we obtain the theoretical maximum discovery latency L corresponding to A. This yields a set of data pair (A, L). Shown in the second step of Figure 4. By collecting such pairs, we construct the "broadcast interval–discovery latency" distribution for a fixed scan

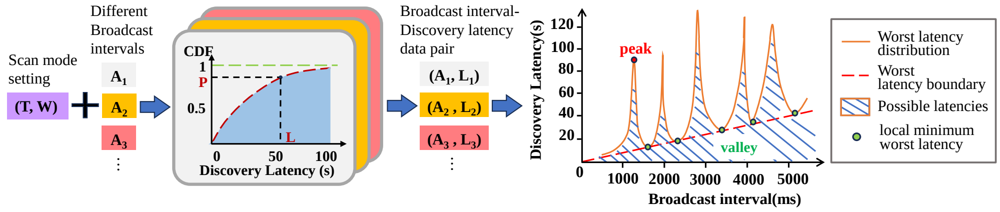


Figure 4: For a specific scan mode (T, W), set an initial broadcast interval value A_1. Gradually increase the value of A and continuously run the BLE NDP simulator to obtain CDF graphs under different (T, W, A) combinations. Select an appropriate P value (typically 90% - 95%) to obtain the corresponding L value. Using the (A, L) sequence, the broadcast interval-discovery latency distribution for this scan mode can be derived.

mode (T, W), illustrated in the final step of Figure 4. In the following, we will briefly refer to it as the A-L Distribution.

As shown in the final step of Figure 4, the A-L Distribution is a waveform with peaks and valleys, where the blue area represents the possible discovery latencies under a certain P value. The orange lines show the distributions of the theoretical worst discovery latencies. In order to achieve the lowest possible discovery latency, when selecting the broadcast interval, we should avoid the intervals leading to peaks and their surrounding areas. The study [4] demonstrates that the local minimum worst discovery latency in this distribution is bounded by an insurmountable limit, which is inversely proportional to the scan mode's duty cycle W/T. This bound corresponds to a straight line passing through the origin point with a positive slope of $P \cdot T/W$. Therefore, along this boundary, points with smaller broadcast intervals exhibit lower discovery latencies.

3. Related Work

The main purpose of OFN is to achieve target positioning. As of now, there have been many works on positioning, such as [19–22]. In addition, in some cases, it is necessary to consider the lack of GPS assistance to achieve positioning function. as described in [23]. During the deployment of OFN, users who carrying the "find devices" are required to upload their location information, which may involve issues of infringing on their privacy. Similar privacy protection in IoT environments has also been studied in [24, 25].

To achieve the shortest possible discovery latency under a given power budget, a wide range of broadcast and scan configuration methods have been proposed for BLE neighbor discovery [4, 5, 26]. For example, the first rigorous duty cycle-dependent bounds on the worst-case discovery latency for deterministic neighbor discovery (ND) protocols are established in [4]. Kindt et al. [26] Analyzes the discovery performance of BLE devices from the perspectives of protocol design and latency bounds, emphasizing that when evaluating the neighbor discovery mechanism, not only should the average latency be considered, but also the worst-case latency and the characteristics of the distribution tail. Through mathematical proof, it demonstrates that no ND protocol can surpass these theoretical bounds. There are also works done for modeling analysis of BLE NDP [9, 27–29]. An analytical model for three-channel-based neighbor discovery in BLE networks is proposed in [28], which inspires subsequent BLE NDP research. A theoretical model based on the Chinese Remainder Theorem (CRT) is proposed in [29], and the discovery latency of BLE NDP is also analyzed using this model in the paper. A BLE neighbor discovery simulator 'Blender' which generates a cumulative distribution function of discovery latency for given parameter configurations involving T, W, and A is given in [9], which inspires the development of our proposed CPBIS mechanism.

Although offline BLE tags have been widely deployed, their current configurations in Offline Finding Network remain relatively simplistic. For example, the broadcast interval of Apple's Air Tag in its lost state is fixed at 2000 ms [30]. This fixed configuration often leads to suboptimal performance in BLE neighbor discovery, manifesting as excessively high discovery latency and a low discovery success rate in practical scenarios. In order to achieve an ideal NDP performance, parameter configurations such as broadcast interval, scan interval, and scan window are particularly important. In the paper [5], after modeling the NDP process, some key parameters in the BLE NDP are given recommended configurations, in addition to the configurations of the three common parameters T, W, and A, the random broadcast delay are also restricted. The article mainly considers the impact of broadcast interval on NDP discovery latency, and finds that there are a large number of peaks in the "broadcast interval-discovery latency" distribution, and it is necessary to be careful to avoid parameters that lead to peaks in the actual parameter configuration.

To optimize discovery latency in the BLE Neighbor Discovery Process, a method to determine the optimal broadcast interval for BLE broadcast devices is proposed in [6]. The goal is to minimize the time for a scan device to detect all surrounding advertisers. However, this study primarily addresses scenarios involving multiple broadcast devices, all of which need to be discovered. In contrast, the OFN typically involves a single broadcast device (i.e., the offline tag), while the number of scan devices is unknown. In this scenario, it only requires that one of the scan devices can detect the offline BLE tag.

To address this particular use case, the ElastiCast broadcast mode, specifically designed for multiple scan modes of varying market share, is proposed [7]. ElastiCast is the first work to systematically address the OFN problem and proposes a corresponding method. ElastiCast includes two broadcast patterns: the Single Broadcast Pattern (ElastiCast-SBP) and the Alternation Broadcast Pattern (ElastiCast-ABP). Specifically, ElastiCast SBP is a single-interval broadcast mode optimized for multiple scan modes. ElastiCast ABP can adopt multiple broadcast intervals alternately and adjust the proportion of each interval. However, this method is still unable to achieve the theoretically lowest discovery latency.

4. CPBIS Mechanism Design

4.1. Key Parameters in CPBIS-mechanism

In this paper, the constraint on the offline BLE tag's broadcast interval represents the restriction on its power consumption. The selection of the broadcast intervals needs to satisfy the minimum equivalent broadcast interval constraint, which means that the equivalent broadcast interval $\overline{A}$ of the selected broadcast intervals cannot be smaller than $A_{\min}$ of the maximum power constraint. The calculation formula for $\overline{A}$ is presented in Equation (1), and the minimum equivalent broadcast interval constraint that it must satisfy is defined in Inequality (2). i is the index of the selected broadcast interval, δ_i is the proportion of the selected broadcast interval in the final broadcast mode (the sum of δ_i is 1), and A_i is the specific value of the i-th broadcast interval.

$$\overline{A} = \sum_{i=1}^{m} \delta_i A_i \tag{1}$$

$$\overline{A} \geq A_{\min} \tag{2}$$

The weighted average discovery latency, is calculated by the following formula.

$$\overline{L} = \sum_{i=1}^{n} \delta_i \left(\sum_{j=1}^{m} \omega_j L_{ij} \right) \tag{3}$$

where, n represents the number of selected broadcast intervals, and m represents the number of existing scan mode types. ω_j is the market share of the j-th scan mode, which needs to be obtained by investigation and serves as an input parameter for CPBIS. It can be realized and periodically updated through industry collaboration (e.g., aggregated statistics from major OFN vendors like Apple/Huawei). The sum of market shares ω_j is 1. L_{ij} is the discovery latency of the i-th broadcast interval under the j-th scan mode in the A-L Distribution. Figure 5 shows the A-L distributions of two scan modes with two broadcast intervals are selected. And the $\overline{L}_{(A_1,A_2)}$ is calculate as $\overline{L}_{(A_1,A_2)} = \delta_{A_1}(\omega_1 \cdot L_{11} + \omega_2 \cdot L_{12}) + \delta_{A_2}(\omega_1 \cdot L_{21} + \omega_2 \cdot L_{22})$. More broadcast intervals or scan modes can be added following the same formula format to incorporate the corresponding parameters for calculating $\overline{L}$.

4.2. CPBIS Mechanism Framework

This section presents the structure of the CPBIS mechanism, outlining the core steps of CPBIS. CPBIS mechanism reduces the discovery latency for finder devices in searching for the offline tag and improves the success rate of this process by optimizing the broadcast mode of the tags. CPBIS analyzes the different scan mode configurations of the known finder devices and calculates the optimal dual-interval broadcast mode for the offline tag.

Figure 6 illustrates the main steps of the CPBIS mechanism. The input consists of the parameter configurations of various scan modes, including the scan interval, scan window, and the market share of each scan mode. The output is a dual-broadcast

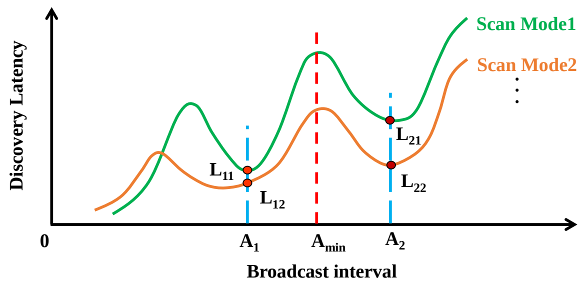


Figure 5: A-L Distributions of two scan modes and two selected broadcast intervals A_1, A_2. L_{11}, L_{21} represent the discovery latencies of A_1, A_2 under scan mode 1, while L_{12}, L_{22} represent the discovery latencies of A_1, A_2 under scan mode 2.

interval broadcast mode composed of the optimal two broadcast intervals with their corresponding proportions.

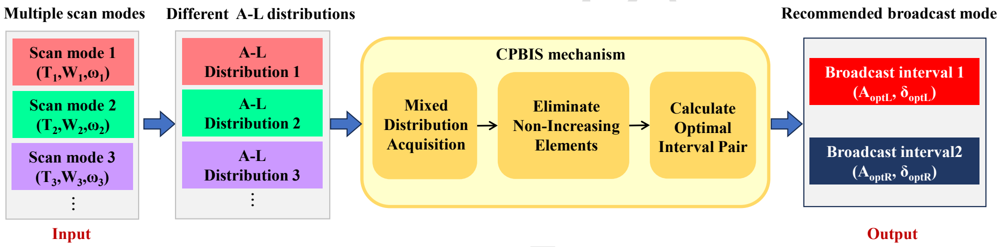


Figure 6: Different scan modes correspond to different A-L distributions. The Mixed Distribution Acquisition module of CPBIS superimposes the A-L distributions of different scan modes according to the market share ω. Eliminate Non-increasing Elements deletes unnecessary points in the mixed distribution. Calculate Optimal Interval Pair figures out the theoretical optimal two broadcast intervals and their proportions, that is, the target optimal two-interval broadcast mode.

Mixed Distribution Acquisition. In this part, the discovery latencies of the A-L distributions of each scan mode obtained in the preliminary preparation work are multiplied by the market share as the weight value and then summed to form a new discovery latency. Since the range of the broadcast intervals on the abscissa is fixed, they will form a new mixed distribution.

Eliminate Non-Increasing Elements. This part defines and removes the non-increasing elements in the mixed distribution diagram, ultimately yielding a strictly monotonically increasing optimized A-L scatter sequence. It also proves that the optimal solution for selecting two broadcast intervals does not lie among these non-increasing points.

Calculate Optimal Interval Pair. Taking $A_{\min}$ as the decision boundary, all the broadcast intervals in the optimized scatter sequence are divided into two sets, the left set and the right set. In addition, a Lower Convex Hull algorithm is provided to select the optimal two broadcast intervals. Then, these two intervals are substituted into Inequality (2), and the limiting case when the equal sign holds is taken as the final broadcast mode.

4.3. Mixed Distribution Acquisition

After obtaining the "broadcast interval-discovery latency" distribution of each scan mode, considering that the market shares of each scan mode are different, the impact degree of the discovery latency of each scan mode on the final weighted average discovery latency also varies. The weighted average discovery latency is an important criterion to judge whether the selection of a broadcast interval is reasonable. Therefore, we superimpose the discovery latency distributions of the required scan modes according to the market shares to form a mixed discovery latency distribution. That is, by multiplying the discovery latency by the market share, all the "broadcast interval-discovery latency" distributions are superimposed on the Y-axis. Consequently, not only is the initial situation taken into account, but also the computational complexity is significantly reduced.

4.4. Eliminate Non-Increasing Elements

First, we define the non-increasing points. Begin the exploration from the second point from the right side of the Mixed Distribution and traverse toward the left. If there exist points with smaller discovery latencies than the currently explored point on the right side of that point, the current point is termed a non-increasing element. All such points should be pruned since the optimal solution cannot reside among them. Based on this, we present the following theorem and its proof.

Theorem 4.1. *When screening for the combination of broadcast intervals that results in the lowest discovery latency, regardless of the combination, non-increasing points cannot be part of the optimal solution.*

Proof. The objective is to identify the combination of broadcast intervals that produces the minimum weighted average discovery latency under the constraint of inequality (2), while also determining the proportion of these intervals. A segment of the A-L distribution is illustrated in Figure 7.

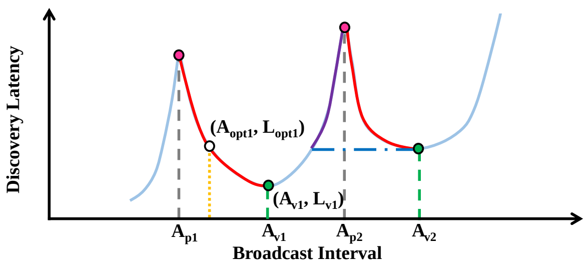


Figure 7: Eliminate non-increasing points, step 1: remove the red part, step 2: remove the purple part.

First, analyze a peak-valley interval $A_{p1} - A_{v1}$. Assume that one of the optimal elements, denoted as (A_{opt1}, L_{opt1}), lies between A_{p1} and A_{v1} (excluding (A_{v1}, L_{v1})) with a proportional weight of δ_{opt1}. While the equivalent broadcast interval of the remaining optimal intervals is $\overline{A}_{opt*}$, the weighted average discovery latency of the remaining optimal intervals is $\overline{L}_{opt*}$. Then, for the equivalent broadcast interval, both have

$$\overline{A}_{opt*} + \delta_{opt1} \cdot A_{opt1} \geq A_{\min}$$

$$\overline{A}_{\text{opt}*} + \delta_{opt1} \cdot A_{v1} \geq A_{\min}$$

For the weighted discovery latency,

$$\overline{L}_{(opt1,opt*)} = \overline{L}_{opt*} + \delta_{opt1} \cdot L_{opt1}$$

$$\overline{L}_{(v1,opt*)} = \overline{L}_{opt*} + \delta_{opt1} \cdot L_{A_{v1}}$$

Since the peak-valley interval $A_{p1} - A_{v1}$ is a monotonically decreasing function, we have

$$\delta_{opt1} \cdot L_{A_{v1}} < \delta_{opt1} \cdot L_{opt1}$$

$$\overline{L}_{(v1,opt*)} < \overline{L}_{(opt1,opt*)}$$

Therefore, the interval $A_{p1} - A_{v1}$(excluding A_{v1}) can be safely pruned, as the optimal solution cannot reside within it—all points in this interval have lower priority than A_{v1}. Similarly, after partitioning the Mixed Distribution into peak-valley-peak segments, the left half of each segment (i.e., the peak-to-valley subinterval, excluding the valley point) can be eliminated for the same reason.

Subsequently, analyze the valley-peak interval $A_{v1} - A_{p2}$. By drawing a horizontal line extending leftward from the valley point A_{v2}, this line intersects the waveform of the $A_{v1} - A_{p2}$ interval at a specific point. Based on the preceding analysis, the portion of the $A_{v1} - A_{p2}$ interval above this horizontal line can be safely deleted, because their priorities are lower than the valley point corresponding to A_{v2}. □

4.5. Calculate Optimal Interval Pair

After superimposing the original A-L distribution plots from various scan modes and pruning non-increasing elements, the resulting sequence is a strictly monotonically increasing A-L scatter plot. At this stage, the complexity of the problem is significantly reduced.

We propose a slope-based solution and an improved lower convex hull algorithm based on this to obtain the optimal dual-interval broadcast mode. This method divides the broadcast intervals into two parts using the minimum equivalent broadcast interval $A_{\min}$, and then determines the optimal two broadcast intervals based on the Lower Convex Hull algorithm. It first locks in the intervals and then calculates their respective proportions to obtain the theoretical optimal broadcast mode.

The specific operation steps of the lower convex hull algorithm are as follows:

- Taking $A_{\min}$ as the dividing line, the (A, L) points in the optimized A–L scatter sequence obtained through **Eliminate Non-Increasing Elements** are divided into two sets: $\mathcal{P}_L$ on the left side of $A_{\min}$ and $\mathcal{P}_R$ on the right side.
- Find the points that form the lower convex hull among all points. The two adjacent points on the hull that span $A_{\min}$ are the optimal points, and their broadcast intervals represent the optimal intervals.
- Substitute the two broadcast intervals selected in the last step into the limiting case when the equal sign in Inequality (2) holds, and obtain the proportion of the two broadcast intervals. The combination of the two broadcast intervals forms the optimal dual-interval broadcast mode.

We will provide a strict proof process for this scheme in the next section **Optimal two intervals selection**, we also prove that the discovery latency generated by an appropriate combination of two broadcast intervals is lower than configurations employing any other number of broadcast intervals.

5. Optimal two intervals selection

After superimposing the A-L distribution diagrams of different scan modes in the **Mixed distribution acquisition** and eliminating the non-increasing elements among them, a strictly monotonically increasing A-L sequence is finally obtained, which we call the optimized A-L scatter sequence. This section elaborates in detail on the method of screening the optimal two broadcast intervals from the optimized A-L scatter sequence based on the slope and convex hull analysis, and provides a mathematical proof of the optimality of the obtained solution.

As discussed before, Let the (A, L) points on the left side of the minimum equivalent broadcast interval $A_{\min}$ (e.g., $A_{\min} = 4000\,\text{ms}$) be denoted as $\mathcal{P}_L$, and the set on the right side as $\mathcal{P}_R$. In Section 5.1, we analyze a specific broadcast interval on one side of $A_{\min}$ and introduce five lemmas to identify its optimal pairing broadcast interval. In Section 5.2, with the final conclusion of the five lemmas, We propose a method based on the Lower Convex Hull to find the optimal broadcast mode.

5.1. Optimal Combination of a Specific Broadcast Interval

We first analyze how to identify the combination with the lowest weighted average discovery latency when pairing a specific point (A_{R1}, L_{R1}) in $\mathcal{P}_R$. We use five lemmas to implement this.

Lemma 5.1. *By selecting one broadcast interval on each side of $A_{\min}$ to form a pair, this pair is guaranteed to satisfy the "minimum equivalent broadcast interval constraint" and will produce a fixed minimum weighted average discovery latency.*

Proof. As shown in Figure 8a, arbitrarily select two intervals A_{L1} in $\mathcal{P}_L$ and A_{R1} in $\mathcal{P}_R$, which are located on the left side and the right side of $A_{\min}$, respectively. The line that passes through two points is denoted as $l : y = kx + b$. (Here, k and b do not refer to specific parameters, but only represent the slope and the intercept of the straight line, and the same applies to the subsequent proofs.)

The minimum equivalent broadcast interval constraint:

$$\alpha A_{L1} + (1-\alpha)A_{R1} \geq A_{\min}$$

Solving for α

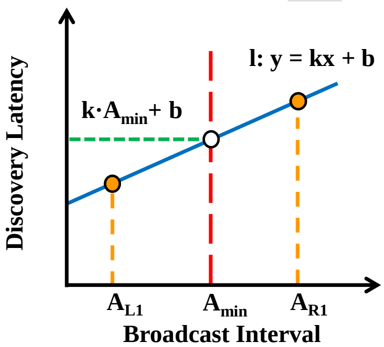
$$0 \leq \alpha \leq \frac{A_{R1} - A_{\min}}{A_{R1} - A_{L1}}$$

The weighted average discovery latency,

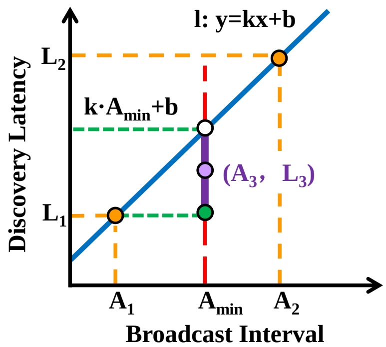
$$\begin{aligned}\overline{L}_{(A_{L1},\ A_{R1})} &= \alpha(kA_{L1} + b) + (1-\alpha)(kA_{R1} + b) \\ &= k[\alpha A_{L1} + (1-\alpha)A_{R1}] + b \\ &\geq kA_{\min} + b\end{aligned}$$

Since the function is monotonically increasing, for any combination of broadcast intervals that satisfies the constraints, when α takes the maximum value, the sum of the weighted average discovery latencies is the smallest, that is, the optimal solution of the discovery latency under this combination is obtained when the equal sign holds.

When $\alpha = \frac{A_{R1}-A_{\min}}{A_{R1}-A_{L1}}$, the equal sign holds, and $\overline{L}_{(A_{L1},\ A_{R1})}$ takes the minimum value of $kA_{\min} + b$. That is, the vertical coordinate of the straight line at $A_{\min}$. $\square$



(a) A random combination of interval A_{L1}, A_{R1} on each side of $A_{\min}$.



(b) In rare cases, the single-interval broadcast mode of A_3 outperforms the dual-interval broadcast composed of A_1 and A_2.

Figure 8: Any combination on both sides of $A_{\min}$ and the situation where single interval broadcast mode is better than dual-interval broadcast mode of this kind of combination.

Lemma 5.2. *Among all pairs of broadcast intervals that satisfy the minimum equivalent broadcast interval constraint, the pair that yields the lowest weighted average discovery latency must be distributed on each side of $A_{\min}$, with one interval smaller and one larger than $A_{\min}$.*

Proof. 1) Assume two arbitrarily selected broadcast intervals A_1 and A_2 are both located on the left side of $A_{\min}$, the corresponding discovery latencies are L_1 and L_2, respectively.

Constraint according to inequality (2):

$$\alpha A_1 + (1-\alpha)A_2 \geq A_{\min}$$

Solving for α, we get:

$$\alpha \leq \frac{A_2 - A_{\min}}{A_2 - A_1} < 0, (A_1 < A_2)$$

$$\alpha \geq \frac{A_2 - A_{\min}}{A_2 - A_1} > 1, (A_1 > A_2)$$

However, by definition $0 \leq \alpha \leq 1$, hence there is no solution in this case.

2) Now assume A_1 and A_2 are both located on the right side of $A_{\min}$. Then:

$$0 \leq \alpha \leq 1 \leq \frac{A_2 - A_{\min}}{A_2 - A_1},\ (A_1 < A_2)$$

$$\frac{A_2 - A_{\min}}{A_2 - A_1} \leq 0 \leq \alpha \leq 1,\ (A_1 > A_2)$$

It is completely feasible, and the corresponding weighted average discovery latency is:

$$\begin{aligned}\overline{L}_{(A_1,A_2)} &= \alpha L_1 + (1-\alpha)L_2 \\ &\geq min\ (L_1|L_2)\end{aligned}$$

3) Now consider an arbitrary interval A_3 on the left side of $A_{\min}$. There exists β such that:

$$\beta A_1 + (1-\beta)A_3 \geq A_{\min}$$

The corresponding weighted average discovery latency is:

$$\overline{L}_{(A_1,A_3)} = \beta L_1 + (1-\beta)L_3 \leq L_1$$

Similarly, when replacing A_1 with A_3, we will get $\overline{L}_{(A_2,A_3)} \leq L_2$. As long as A_3 is combined with the smaller one of A_1 and A_2, the minimum discovery latency produced will be smaller than that of the combination of A_1 and A_2. This shows even though two intervals both on the right side of $A_{\min}$ satisfy the minimum equivalent broadcast interval constraint, they result in a higher discovery latency than a pair of intervals distributed on each side of $A_{\min}$.

Therefore the two optimal broadcast intervals must be distributed on either side of $A_{\min}$. □

The original proposition is proved. Similarly, this also proves that in most cases, the weighted average discovery latency generated by an appropriate combination of two broadcast intervals is lower than that of a single broadcast interval. Only in rare cases does the dual-interval broadcast mode degenerate into a single-interval broadcast mode. Suppose that A_1 and A_2 are the final solutions obtained. Only when there is a point (A_3, L_3) at the position of $A_{\min}$, and its discovery latency is within the interval $[L_1, k \cdot A_{\min} + b)$ can the single interval be superior to the dual interval, shown in Figure 8b. We include this situation in the introduction of the convex hull algorithm in the section 5.2.

In fact, this situation can only occur when there happens to be an optional point at $A_{\min}$. Due to the existence of the worst latency boundary and the monotonically increasing characteristic of the optimized A-L scatter point sequence, this possibility is further reduced.

In addition, the study in [7] points out that the optimal solution under the single broadcast interval mode will change due to the adv delay (that is, after adding the adv delay to the optimal single broadcast interval, the value becomes larger and moves away from the valley), while the dual broadcast intervals can effectively improve this situation.

Lemma 5.3. *Under the condition that **Lemma 5.2** is satisfied, for any combination of two broadcast intervals distributed on both sides of $A_{\min}$ on the same straight line $l : y = kx + b$, the minimum weighted average discovery latency generated is a fixed value.*

Proof. As shown in Figure 9a, suppose there are two combinations, (A_{L2}, A_{R2}) and (A_{L3}, A_{R3}), where A_{L2} and A_{L3} are located to the left of $A_{\min}$, and A_{R2} and A_{R3} are located to the right of $A_{\min}$, along the same straight line. Both combinations satisfy the minimum equivalent broadcast interval constraint $\overline{A} \geq A_{\min}$. The weighting proportion of A_{L2} is γ (and A_{R2} is $1-\gamma$); similarly, the proportion of A_{L3} is η (and A_{R3} is $1-\eta$).

According to the minimum equivalent broadcast interval constraint:

$$\begin{cases} \gamma A_{L2} + (1-\gamma)A_{R2} \geq A_{\min} \\ \eta A_{L3} + (1-\eta)A_{R3} \geq A_{\min} \end{cases}$$

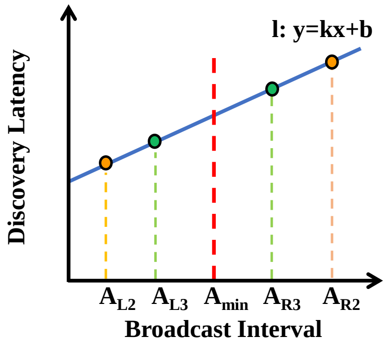
For the combination (A_{L2}, A_{R2}), $(\gamma, 1-\gamma)$,

$$\begin{aligned} &\textit{discovery latency } \overline{L}_{(A_{L2},A_{R2})} \\ &= \gamma(kA_{L2} + b) + (1-\gamma)(kA_{R2} + b) \\ &= k[\gamma A_{L2} + (1-\gamma)A_{R2}] + b \\ &\geq kA_{\min} + b \end{aligned} \tag{4}$$

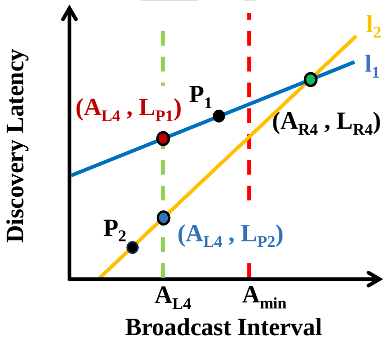
In the same way,combination (A_{L3}, A_{R3}), $(\eta, 1-\eta)$,

$$\begin{aligned} &\textit{discovery latency } \overline{L}_{(A_{L3},A_{R3})} \\ &= \eta(kA_{L3}+b)+(1-\eta)(kA_{R3}+b) \\ &= k[\eta A_{L3}+(1-\eta)A_{R3}]+b \\ &\geq kA_{\min}+b \end{aligned} \tag{5}$$

The minimum values of equations (4) and (5) are equal, confirming the validity of the original proposition. □



(a) Two combinations (A_{L2}, A_{R2}) and (A_{L3}, A_{R3}) on the same straight line l.



(b) Two straight lines passing through the point (A_{R4}, L_{R4}).

Figure 9: Analysis of Different Combinations on Both Sides of $A_{\min}$.

Lemma 5.4. *For the set $\mathcal{P}_R$, consider any point (A_{R4}, L_{R4}). Among all the combinations of this point with points in the set $\mathcal{P}_L$, the combination with the largest slope corresponds to the minimum weighted average discovery latency.*

Proof. Shown as Figure 9b, assume that there exist two points P_1 and P_2 in the set $\mathcal{P}_L$. Draw lines l_1 and l_2 through the point (A_{R4}, L_{R4}) passing through P_1 and P_2 respectively. (Since the optimized A-L scatter sequence is a strictly monotonically increasing sequence, the slopes of these two lines must both be positive.)

Additionally, select an arbitrary abscissa A_{L4} on the left side of $A_{\min}$. The coordinates of the intersection points of A_{L4} with the two lines are (A_{L4}, L_{p1}) and (A_{L4}, L_{p2}), respectively. Whether the points (A_{L4}, L_{p1}) and (A_{L4}, L_{p2}) are in the set A_L does not affect the subsequent judgment.

1). Slope comparison:

$$l_1 : k1 = \frac{L_{R4} - L_{P1}}{A_{R4} - A_{L4}}$$

$$l_2 : k2 = \frac{L_{R4} - L_{P2}}{A_{R4} - A_{L4}}$$

$$k1 - k2 = \frac{L_{P2} - L_{P1}}{A_{R4} - A_{L4}}$$

2). comparison of sum of weighted average discovery latency:

Suppose that for the combination (A_{L4}, A_{R4}), there exists a proportion $(\epsilon, 1-\epsilon)$ such that the constraint $\overline{A} \geq A$ is satisfied, then:

$$for\ l_1 : \overline{L}_{(P_1,A_{R4})} = \epsilon L_{p1} + (1-\epsilon) L_{R4}$$
$$for\ l_2 : \overline{L}_{(P_2,A_{R4})} = \epsilon L_{p2} + (1-\epsilon) L_{R4}$$
$$\overline{L}_{(P_1,A_{R4})} - \overline{L}_{(P_2,A_{R4})} = \epsilon (L_{p1} - L_{p2})$$

If $L_{p1} > L_{p2}$, then $\overline{L}_{(P_1,A_{R4})} > \overline{L}_{(P_2,A_{R4})}$ and $k_1 < k_2$. The combination with the steeper slope yields a lower weighted average discovery latency at $A_{\min}$, and vice versa. According to **Lemma 5.3**, the original proposition is proved. □

Similarly, we obtain **Lemma 5.5**.

Lemma 5.5. *For the set $\mathcal{P}_L$, consider any point in $\mathcal{P}_L$. Among all combinations of this point with the points in the set $\mathcal{P}_R$, the combination with the lowest slope corresponds to the minimum weighted average discovery latency.*

lemma 5.4 and **lemma 5.5** are derived under the extreme condition (when the constraint of the minimum equivalent broadcast interval is an equality). These two lemmas can be used to verify whether the obtained solution is the optimal dual-interval solution.

Theorem 5.1. *If there exist two intervals A_{optL} and A_{optR} on both sides of $A_{\min}$ that satisfy **Lemma 5.4** and **Lemma 5.5**, then A_{optL} and A_{optR} are the optimal two intervals that can produce the minimum weighted average discovery latency.*

Proof. We use proof by contradiction to demonstrate this:

1). For the two broadcast intervals A_{optL} and A_{optR}, with $A_{\min}$ as the boundary, this combination forms the line with the maximum slope among all combinations between A_{optR} and the points in $\mathcal{P}_L$.

if there exists a point (A_{pl1}, L_{pl1}) on the left of $A_{\min}$ that lies below the line segment between (A_{optL}, L_{optL}) and (A_{optR}, L_{optR}), then the slope of the line through (A_{pl1}, L_{pl1}) and (A_{optR}, L_{optR}) is strictly greater than the slope of the original pair (A_{optL}, L_{optL}) and (A_{optR}, L_{optR}).

Let

$$s_{optL,optR} := \frac{L_{optR} - L_{optL}}{A_{optR} - A_{optL}}, \qquad s_{pl1,optR} := \frac{L_{optR} - L_{pl1}}{A_{optR} - A_{pl1}}.$$

Assume that $A_{optL} < A_{pl1} < A_{optR}$, and that the point (A_{pl1}, L_{pl1}) lies strictly below the line segment formed by (A_{optL}, L_{optL}) and (A_{optR}, L_{optR}), i.e.,

$$L_{pl1} < L_{optL} + (A_{pl1} - A_{optL})\, s_{optL,optR}.$$

Rearranging gives

$$L_{optR} - L_{pl1} > L_{optR} - \big(L_{optL} + (A_{pl1} - A_{optL})s_{optL,optR}\big).$$

Since $L_{optR} - L_{optL} = (A_{optR} - A_{optL})s_{optL,optR}$, the right-hand side simplifies to

$$L_{optR} - L_{pl1} > s_{optL,optR}\big((A_{optR} - A_{optL}) - (A_{pl1} - A_{optL})\big) = s_{optL,optR}(A_{optR} - A_{pl1}).$$

Because $A_{optR} - A_{pl1} > 0$, dividing both sides yields

$$\frac{L_{optR} - L_{pl1}}{A_{optR} - A_{pl1}} > s_{optL,optR}.$$

That is,

$$s_{pl1,optR} > s_{optL,optR}.$$

Therefore, there is no point in $\mathcal{P}_L$ to the left of $A_{\min}$ that lies below the extended line between $(A_{\mathrm{optL}}, L_{\mathrm{optL}})$ and $(A_{\mathrm{optR}}, L_{\mathrm{optR}})$. Otherwise, A_{optR} could form a combination with it with a larger slope, thus violating **Lemma 5.4**.

Similarly, the same is true when $A_{pl1} < A_{optL} < A_{optR}$.

2). Similarly, with $A_{\min}$ as the boundary, this combination also forms the line with the minimum slope among all combinations between A_{optL} and the points in $\mathcal{P}_R$. There is no point in $\mathcal{P}_R$ to the right of $A_{\min}$ that lies below the extended line between $(A_{\mathrm{optL}}, L_{\mathrm{optL}})$ and $(A_{\mathrm{optR}}, L_{\mathrm{optR}})$.

It can be observed that the selected combination forms a "boundary," below which no other (A, L) points exist. Since all remaining (A, L) points lie above the extension of this line segment, the weighted average latency corresponding to the interpolation of this combination at $A_{\min}$ is the smallest. Therefore, if an optimal pair of broadcast intervals exists, they must lie on the lower convex hull of all (A, L) points. By substituting these two intervals into the minimum equivalent broadcast interval constraint and solving for the equality case, the optimal proportion can be obtained. In rare cases, the dual-interval mode degenerates into a single-interval mode, as discussed in **Lemma 5.2**. □

5.2. Slope-Based Screening strategy and Lower convex hull algorithm for Optimal Dual-Broadcast Intervals

5.2.1. Slope-Based Screening strategy for Optimal Dual-Broadcast Intervals

Based on the **Lemma 5.4** and **Theorem 5.1** from the previous section, we perform the following operation on the filtered segmented broadcast interval sequence:

- Starting from the first point in the set P_R located to the right of $A_{\min}$, we sequentially select the points that yield the largest slope, thereby constructing an (A, L) sequence guaranteed to contain the optimal solution. Finally, the combination with the smallest discovery latency within this sequence is chosen as the optimal solution. The correctness of this procedure can be justified by **Lemma 5.4** and **Theorem 5.1**.

- Since the selected intervals are determined, the proportions of the two intervals are also fixed at this time. Substitute the broadcast interval values of the two points into Inequality (2) and solve for the proportions when the equality holds at the limit.

The slope-based strategy requires multiple passes through the (A, L) sequence, and the complexity is $O(n^2)$. In the proof of **Theorem 5.1**, we mention that there are no other points below the straight line formed by $(A_{\text{optL}}, L_{\text{optL}})$ and $(A_{\text{optR}}, L_{\text{optR}})$. This implies that the optimal solution lies on the lower convex hull of the entire (A, L) sequence.

5.2.2. Optimality Proof Based on the Lower Convex Hull

Mathematical Modeling of the Problem. In **Eliminate Non-Increasing Elements**, we obtain a strictly monotonically increasing optimized A-L scatter point sequence. At this point we can describe the problem as a linear programming problem.

Given a set of points $\{(A_i, L_i)\}_{i=1}^{n}$, consider the following optimization problem (P):

$$\min_{\delta} \sum_{i=1}^{n} L_i \delta_i \tag{6}$$

$$\text{s.t.} \sum_{i=1}^{n} \delta_i = 1, \quad \sum_{i=1}^{n} A_i \delta_i = A_{\min}, \quad \delta_i \geq 0 (i = 1, \ldots, n). \tag{7}$$

This problem aims to minimize the weighted discovery latency subject to convex combination constraints and a prescribed average broadcast interval $A_{\min}$.

Step 1 — At Most Two Points are Needed. Problem (P) is a linear program with n variables and two equality constraints. By the basic feasible solution (BFS) property of linear programming, for a linear program with m independent equality constraints, any basic feasible solution (including the optimal solution) has at most m positive variables (i.e., non-zero δ_i) in non-degenerate cases. Therefore, there are at most 2 non-zero δ_i in the optimal solution, that is, the optimal solution is supported by at most 2 points. Equivalently, if $A_{\min}$ coincides with some A_k, the optimal solution is the single point (A_k, L_k) (The exception discussed in Figure 8b in **Lemma 5.2**). Otherwise, the optimum lies on a line segment between two points (A_i, L_i) and (A_j, L_j), $i < j$, with weights(When the weighted average broadcast interval is exactly equal to $A_{\min}$)

$$\delta_i = \frac{A_j - A_{\min}}{A_j - A_i}, \qquad \delta_j = \frac{A_{\min} - A_i}{A_j - A_i}.$$

The resulting latency is the linear interpolation

$$L_{i,j}(A_{\min}) = \delta_i L_i + \delta_j L_j = L_i + (A_{\min} - A_i)\frac{L_j - L_i}{A_j - A_i}.$$

That is, the vertical coordinate of the line segment $\overline{(A_i, L_i)(A_j, L_j)}$ evaluated at $A = A_{\min}$. The global optimal discovery latency among all legal pairings is the minimum height L at the horizontal coordinate $A_{\min}$ on all line segments (connecting any left and right points of $A_{\min}$).

Step 2 — Convex Hull and Lower Convex Hull. Define the convex hull

$$\mathrm{conv}(\{(A_i, L_i)\}) = \left\{(A, L) : (A, L) = \sum_{i=1}^{n} \delta_i (A_i, L_i), \sum \delta_i = 1, \ \delta_i \geq 0\right\}.$$

For any fixed horizontal coordinate A, the minimum achievable vertical coordinate is

$$f(A) = \min\{L : (A, L) \in \mathrm{conv}(\{(A_i, L_i)\})\}.$$

This function $f(A)$ is precisely the lower convex hull of the point set $conv(\{(A_i, L_i)\})$. By definition, for any convex combination satisfying $\sum_{i=1}^{n} \delta_i = 1, \sum_{i=1}^{n} A_i \delta_i = A$, the corresponding vertical coordinate $\sum_{i=1}^{n} L_i \delta_i$ is always no smaller than $f(A)$. The objective of (P) is precisely to figure out $f(A_{\min})$.

In the slope-based screening strategy of Section **5.2.1**, for each point p_r to the right of $A_{\min}$ we compute the maximum slope s_{p_l,p_r} over all points p_l to the left of $A_{\min}$, and then obtain the corresponding value $L_{p_l^*,p_r}(A_{\min})$. The final solution is given by the minimum of these values across all p_r. This procedure is equivalent to enumerating all possible line segments and selecting the one with the smallest latency, which requires $O(n^2)$ time in the worst case since each p_r must scan all candidate points p_l.

The lower convex hull reduces this exhaustive search to $O(n)$ by exploiting geometric monotonicity. Specifically, the hull preserves only those points p_l that can potentially yield the maximum slope for some p_r, thereby eliminating redundant candidates. As a result, it is unnecessary to re-scan the entire set of left points for each p_r.

Step 3 — Two-Point Interpolation Corresponds to Hull Edges of Lower Convex Hull. According to Step 1, the optimal solution of problem (P) is given by either a single point (A_i, L_i) or a pair of points (A_i, L_i) and (A_j, L_j). The objective value is equal to the vertical coordinate of the line connecting these two points at $A = A_{\min}$, denoted as $h_{i,j}(A_{\min})$ (previously referred to as $L_{i,j}(A_{\min})$).

Any feasible interpolation between two points with indices i and j defines a line

$$h_{i,j}(A) = L_i + (A - A_i)\frac{L_j - L_i}{A_j - A_i}.$$

Over the interval $[A_i, A_j]$, this line always satisfies $h_{i,j}(A) \geq f(A)$, since the lower envelope $f(A)$ represents the minimal vertical coordinate attainable by convex combinations. Equality holds if and only if the segment $\overline{(A_i, L_i)(A_j, L_j)}$ lies on the lower convex hull. Consequently, the global optimum at $A = A_{\min}$ is achieved exclusively by the hull edge that spans $A_{\min}$(i.e., the two adjacent vertices on the lower convex hull that span across $A_{\min}$). Any other feasible convex combination yields a strictly higher latency value at $A_{\min}$.

Step 4 — Algebraic Contradiction if the optimal Segment is Not on the Hull. For contradiction, Assume that the optimal solution uses two points with indices i and j whose line segment is not part of the lower hull. Then there exists a third point (A_k, L_k) with $A_i < A_k < A_j$ and $L_k < h_{i,j}(A_k)$. The piecewise linear path through (i, k) and (k, j) lies strictly below $h_{i,j}$ on $[A_i, A_j]$. Therefore at $A_{\min}$, at least one of the sub-segments gives a strictly smaller value than $h_{i,j}(A_{\min})$, contradicting the assumed optimality of (i, j). Thus the optimal line segment must coincide with an edge of the lower convex hull.

Special and Degenerate Cases..

- If $A_{\min} = A_k$ is exactly equal to the horizontal coordinate of a point on the lower convex hull, then the optimality can be achieved by this single point. As discussed in **Lemma 5.2** and Figure 8b.
- If multiple points are collinear on the lower convex hull, any convex combination of adjacent collinear vertices yields the same optimal value.

The above analysis establishes that the global optimal solution to this problem is realized by the edge of the *lower convex hull* that spans the vertical line $A = A_{\min}$. The two endpoints of this edge, together with their interpolation weights, yield the optimal convex combination and the minimum achievable weighted average discovery latency.

The algorithm of the Lower Convex Hull is like Algorithm 1. The time complexity of the entire algorithm is T(n) = O(mn) (m: number of scan modes, n: number of broadcast intervals), the space complexity is S(n) = O(mn). For the **Mixed Distribution Acquisition** part. Since it is required to superimpose n discovery latencies from A-L distribution of the m scan modes, the computational complexity of this module is: T (n) = O (mn), and S (n) = O (mn), see lines 1-3 of **Algorithm 1**. For the **Eliminate Non-Increasing Elements**, only a single traversal is required, yielding T (n) = O (n), S (n) = O (n), see lines 4-8 of **Algorithm 1**. For the **Calculate Optimal Interval Pair**, the computational complexity is also T (n) = O (n)(Constructing the lower convex hull is O(n), searching the interval is O(n)), S (n) = O (n), see the rest of the code in **Algorithm 1**. Which means regardless of the number of scan modes, the problem size increases only in the first module, it remains unchanged in the latter two modules. The CPBIS algorithm runs on a backend server or computer, and the results are then deployed onto the BLE chip.

6. Performance evaluation

6.1. Experiemntal deployments

In order to experimentally verify our CPBIS-mechanism, we made the following deployments.

We conduct both simulation and real-world experiments. In the simulation experiments, we compare the weighted average discovery latency of each broadcast mode

Algorithm 1: CPBIS algorithm: Find Optimal Broadcast-Interval Data Pair

Input: Interval sequence $A[\,]$; latency sequences $L_1[\,], L_2[\,], \ldots$; market shares $\omega_1, \omega_2, \ldots$; minmum equivalent broadcast interval$A_{\min}$
Output: The optimal dual-interval broadcast mode: $(A_{\text{optL}}, \delta_{\text{optL}})$ and $(A_{\text{optR}}, \delta_{\text{optR}})$

for $i \leftarrow 0$ **to** $len(A) - 1$ **do**
    $\overline{L}[i] \leftarrow \sum_{k=1}^{m} \omega_k L_k[i]$;
end
for $idx \leftarrow len(\overline{L}) - 1$ **to** 0 **do**
    if $\overline{L}[idx]$ *is non-increasing* **then**
        delete $\overline{L}[idx]$, delete $A[idx]$;
    end
end
Function *Cross(o,a,b)***:**
    return $(a_0 - o_0) \cdot (b_1 - o_1) - (a_1 - o_1) \cdot (b_0 - o_0)$;
end
$points \leftarrow \text{zip}(A, \overline{L})$;
$lower \leftarrow \emptyset$;
foreach $p \in points$ **do**
    while $|lower| \geq 2$ ***and*** $Cross(lower[-2], lower[-1], p) \leq 0$ **do**
        pop($lower$);
    end
    append($lower, p$);
end
$chosen\_pair \leftarrow$ null;
for $i \leftarrow 0$ **to** $|lower| - 2$ **do**
    $(A_{h1}, L_{h1}) \leftarrow lower[i]$;
    $(A_{h2}, L_{h2}) \leftarrow lower[i+1]$;
    if $A_{h1} \leq A_{\min} \leq A_{h2}$ **then**
        $chosen\_pair \leftarrow ((A_{h1}, L_{h1}), (A_{h2}, L_{h2}))$;
        break;
    end
end
if $chosen\_pair = null$ **then**
    error("Input $A_{\min}$ is out of the valid interval range.");
end
$\delta_{\text{optL}} \leftarrow (A_{h2} - A_{\min})/(A_{h2} - A_{h1})$;
$\delta_{\text{optR}} \leftarrow 1 - \delta_{\text{optL}}$;
$L_{\text{opt}} \leftarrow \delta_{\text{optL}} L_{h1} + \delta_{\text{optR}} L_{h2}$;
return $A_{\text{optL}},\ \delta_{\text{optL}},\ A_{\text{optR}},\ \delta_{\text{optR}},\ L_{\text{opt}}$;

under a fixed market share of scan modes with varying $A_{\min}$ values. We also analyze the weighted average discovery latency by adjusting the market share of scan modes with a fixed $A_{\min}$. In the real-world experiments, we conduct three batches of BLE neighbor discovery tests. The first two batches were carried out in environments with weak BLE interference, where 10–20 Bluetooth signals were detected simultaneously in the surrounding area. Since ElastiCast [7] is the first work to systematically address the OFN problem and proposes a corresponding solution, it does not provide a detailed algorithm for ElastiCast-ABP but only presents parameter configurations for a specific scenario. Therefore, in the first batch, we perform performance comparisons within the same static market environment as in [7] . The third batch was conducted in a typical office scenario with strong interference, where 40–50 Bluetooth signals were detected simultaneously. We compared the weighted average discovery latency of different methods across these scenarios. Notably, in the third batch, all methods except CPBIS exhibited a significant decline in discovery success rates.

Currently, we do not have cooperation with relevant manufacturers. However, the core of performance evaluation lies in BLE neighbor discovery. The real-world experiment is based on the Nordic nRF52832 chip, which is a typical BLE chip. We chose Keil uvision5 [31] as the hardware code editing platform. Adjustments to both the broadcast mode and scan mode of the nRF52832 chips are realized using this platform. We use the nRF5 SDK 17.0.2 [32] officially provided by Nordic and modify it to realize the required functions. The protocol stack is burned into the chip using nRFgo studio [33] provided by Nordic. We use J-Link V7.96 [34] and a J-link emulator (The orange part in the lower right corner in Figure 10) to program and debug. As the J-link RTT Viewer can not save scan logs in real time, it periodically clears data. We have identified the Real Time Transfer Tool (RTT-T) [35] to preserve chip logs.

The BLE neighbor discovery experiment requires a "Scanner" and a "Advertiser". We chose two nRF52832 development boards(The parts marked by green and red circles in Figure 10) as the Advertiser and the Scanner, respectively. The Advertiser simulates the lost BLE tag, and the Scanner simulates the finder device. The nRF52832 chip is chosen because the Bluetooth module used in a real prototype Air Tag is nRF52832. By adjusting the broadcast mode of the chip, we simulate the situation that the offline BLE tag broadcasts its signal to the surroundings when it is lost. The broadcast modes we set include single broadcast interval, two broadcast intervals with the same proportion, and the dual-interval broadcast mode obtained by CPBIS. As for the Scanner, we add a scan report event to it. The event includes the names of the broadcast devices received, enabling us to search the logs and determine if the Advertiser was successfully found. The experimental deployment is shown in Figure 10.

In reality, since the situations of people passing by the offline tag are random, a finder device may enter the tag's broadcast range at any stage of the broadcast. Simulating pedestrians repeatedly entering the broadcast range of the tag in experiments would be time-consuming, and the broadcast range of the tag is not a precise value, which could introduce significant error. Therefore, the experimental proce-

Figure 10: The Advertiser continuously broadcasts after setting a broadcast mode, randomly activates the Scanner to start a BLE neighbor discovery. The Host Device powers both the Advertiser and the Scanner and utilizes J-Link to record the Scanner's scan logs.

dure is as follows. We limit the duration of one BLE neighbor discovery process and ensure the Scanner can capture the Advertiser's broadcast signal within coverage. The Scanner is then activated randomly to simulate pedestrians entering the tag's broadcast range randomly. We record the discovery success rate, calculate the final weighted average discovery latency, and compare the results of different broadcast modes.

We assume three environments:

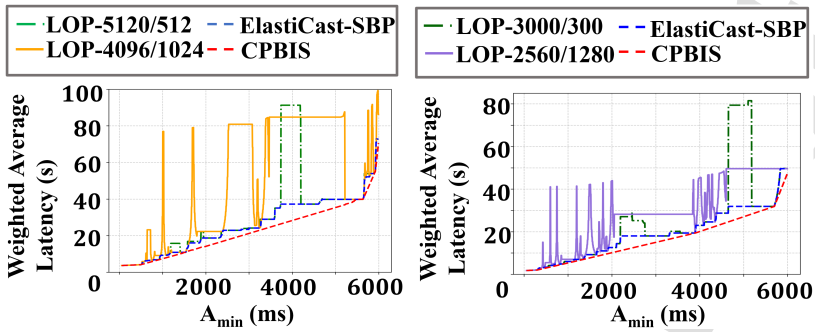
- **Environment 1**: There are two scan modes. Scan Mode 1 has a duty cycle of $T/W = 5120\,\text{ms}/512\,\text{ms}$ with a market share of 0.5, and Scan Mode 2 has $T/W = 4096\,\text{ms}/1024\,\text{ms}$ with a market share of 0.5. The environment contains 10–20 concurrent BLE signals.

- **Environment 2**: Scan Mode 1 has $T/W = 3000\,\text{ms}/300\,\text{ms}$ with a market share of 0.4, and Scan Mode 2 has $T/W = 2560\,\text{ms}/1280\,\text{ms}$ with a market share of 0.6. With 10-20 BLE signals. The environment contains 10–20 concurrent BLE signals.

- **Environment 3**: Scan Mode 1 has a duty cycle of $T/W = 5120\,\text{ms}/512\,\text{ms}$ with a market share of 0.5, and Scan Mode 2 has $T/W = 4096\,\text{ms}/1024\,\text{ms}$ with a market share of 0.5. In a high-interference scenario with 40-50 BLE signals.

6.2. Simulation experiments

For the simulation experiments. We compare the weighted average discovery latencies of different broadcast modes under different minimum equivalent broadcast interval constraints in **Environment 1** and **Environment 2**. For instance, **Environment 1** includes the local optimal broadcast mode LOP 5120/512 optimized for 5120ms/512ms and the local optimal broadcast mode LOP 4096/1024 optimized for 4096ms/1024ms. There is also the ElastiCast-SBP broadcast mode in Paper [7]. The dual-interval broadcast mode of CPBIS-mechanism. As shown in Figure 11, regardless of the environment, the theoretical weighted average discovery latency



(a) Environment 1.

(b) Environment 2.

Figure 11: Theoretical weighted average discovery latency of different broadcast modes in two environments.

achieved by the CPBIS mechanism consistently remains the lowest among the four broadcast modes.

We also adjust the market shares of different scan modes and compare the weighted average discovery latency of each broadcast mode under various market share settings with a fixed A_{min}. As shown in Figure 12 and Figure 13, the results obtained by the CPBIS mechanism remain consistently at the lowest level.

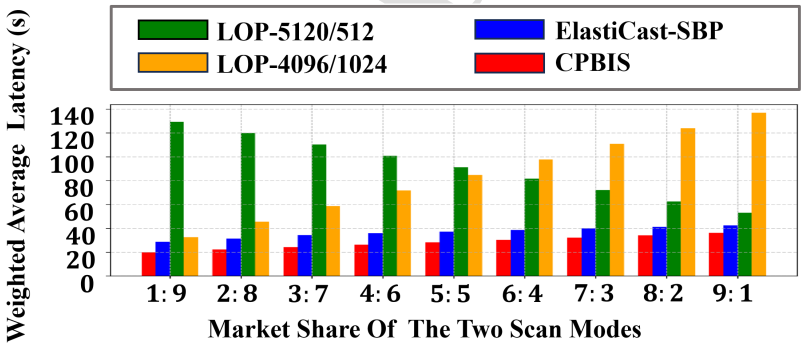


Figure 12: With $A_{\text{min}} = 4000\,\text{ms}$, the market share of scan modes 5120ms/512ms and 4096ms/1024ms is adjusted from 1:9 to 9:1.

6.3. Real-world experiments

In terms of the real-world experiments. For **Environment 1** and **Environment 3**, we set $A_{\text{min}} = 4000\,\text{ms}$ as shown in Figure 12, and the result obtained by CPBIS is 1535 ms with a proportion of 0.3988 and 5635 ms with a proportion of 0.6012. To facilitate the experiment, we adjusted the parameters as follows: in the first stage, 1535 ms with a proportion of 0.4; in the second stage, 5645 ms with a proportion of

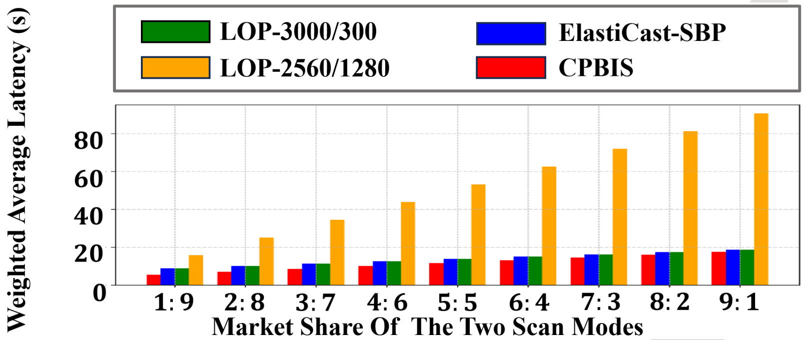


Figure 13: With $A_{\min} = 2000\,\text{ms}$, the market share of scan modes 3000ms/300ms and 2560ms/1280ms is adjusted from 1:9 to 9:1.

0.6. We limit the time of a neighbor discovery to 40 seconds and set the parameters as shown in Table 1 according to each broadcast mode. We also added ElastiCast-ABP to this batch of experiments for comparison.

For **Environment 2**, as shown in Figure 13, we set $A_{\min} = 2000\,\text{ms}$ and limit each discovery attempt to 30 s. LOP 3000/300 employs a single-interval broadcast at 2100 ms. LOP 2560/1280 uses a single-interval broadcast at 2000 ms. ElastiCast-SBP adopts a single-interval broadcast at 2100 ms. CPBIS uses 1350 ms with a proportion of 0.7 and 3900 ms with a proportion of 0.3, featuring a first-stage broadcast duration of 21 seconds and a second-stage broadcast duration of 9 seconds. As shown in the Table 2. As Paper [7] does not provide a detailed algorithm for ElastiCast-ABP but only specifies a parameter configuration for a particular scenario, the parameter settings of ElastiCast-ABP are not included in **Environment 2**.

Table 1: Parameter settings in Environment 1 and Environment 3.

broadcast interval / broadcast mode	the first interval	the second interval
LOP 5120/512	4185ms for 40s	N/A
LOP 4096/1024	5115ms for 40s	N/A
ElastiCast-SBP	4600ms for 40s	N/A
ElastiCast-ABP	2980ms for 20s	5620ms for 20s
CPBIS	1535ms for 16s	5645ms for 24s

In actual operation, we take 10 BLE neighbor discovery processes as a small group and keep increasing the number of small groups until the discovery success rate stabilizes.

The final experimental results for each environment are as follows.

For **Environment 1**, in the 4096ms/1024ms scan mode, each broadcast mode performs 30 neighbor discoveries respectively. In the 5120ms/ 512ms scan mode,

Table 2: Parameter settings in Environment 2.

broadcast interval / broadcast mode	the first interval	the second interval
LOP 3000/300	2100ms for 30s	N/A
LOP 2560/1280	2000ms for 30s	N/A
ElastiCast-SBP	2100ms for 30s	N/A
CPBIS	1350ms for 21s	3900ms for 9s

LOP 5120/512 and LOP 4096/1024 perform 30 neighbor discoveries respectively, ElastiCast-SBP performs 60 times, and ElastiCast-ABP and CPBIS perform 80 random neighbor discoveries respectively. The final results are shown in Table 3. CPBIS reduces the weighted average discovery latency by 31.3% compared with ElastiCast-SBP, while improving the discovery success rate by 7.3%. Compared with ElastiCast-ABP, CPBIS achieves an 11.2% higher discovery success rate.

Table 3: Discovery latency and discovery success rate of different broadcast modes in Environment 1.

scan mode:T/W / broadcast mode	5120ms/512ms	4096ms/1024ms	weighted average latency	weighted average success rate
LOP 5120/512	16.576s	timeout	N/A	46.7%
LOP 4096/1024	timeout	10.337s	N/A	70%
ElastiCast-SBP	17.322s	13.273s	15.298s	90.8%
ElastiCast-ABP	14.304s	7.091s	10.698s	86.9%
CPBIS	14.406s	6.627s	10.517s	98.1%

For **Environment 2** and **Environment 3**, each broadcast mode performs 30 neighbor discoveries under every scan mode. The corresponding results are summarized in Table 4 and Table 5, respectively. In **Environment 2**, LOP 3000/300 and ElastiCast-SBP share identical broadcast configurations, so their results are listed in the same row. Among all 60 tests in this environment, the CPBIS mechanism achieves the lowest weighted average discovery latency of 5.302s. All BLE neighbor discovery procedures are completed successfully, yielding a total weighted average discovery success rate of 100%, which is the highest among all methods.

Table 4: Discovery latency and discovery success rate of different broadcast modes in Environment 2.

scan mode:T/W / broadcast mode	3000ms/300ms	2560ms/1280ms	weighted average latency	weighted average success rate
LOP 2560/1280	timeout	2.840s	N/A	58.3%
LOP 3000/300, ElastiCast-SBP	11.267s	2.633s	6.087s	98.3%
CPBIS	10.257s	1.998s	5.302s	100%

Environment 3 is conducted in a real office scenario with dense coexisting BLE nodes. The weighted average discovery latency produced by CPBIS is 14.548s,

Table 5: Discovery latency and discovery success rate of different broadcast modes in Environment 3.

scan mode:T/W / broadcast mode	5120ms/512ms	4096ms/1024ms	weighted average latency	weighted average success rate
LOP 5120/512	timeout	timeout	N/A	41.6%
LOP 4096/1024	timeout	14.111s	N/A	45%
ElastiCast-SBP	timeout	17.565s	N/A	46.7%
ElastiCast-ABP	timeout	9.951s	N/A	51.7%
CPBIS	18.714s	10.382s	14.548s	83.3%

which is only 4.031s higher than that in **Environment 1** (from 10.517s to 14.548s) under the same scan mode configuration. In this high-interference scenario, all other methods suffer from drastically reduced discovery success rates (below 50%) under the 5120ms/512ms scan mode compared with **Environment 1**. Among non-CPBIS methods, LOP 5120/512 performs best with a success rate of 43.3%. In contrast, CPBIS maintains a success rate of 66.7% under this scan mode and achieves an overall weighted average discovery success rate of 83.3%. This indicates that CPBIS exhibits strong anti-interference capability in real multi-node BLE network topologies.

7. Conclusion

This paper proposes CPBIS, a novel dual-interval broadcast mode screening scheme for BLE offline tags in the offline finding network. The broadcast setting generated by CPBIS achieves low discovery latency for various scan modes, and improves the success rate of BLE neighbor discovery. We prove that CPBIS achieves the theoretically minimum discovery latency for BLE neighbor discovery in OFN and provide a method for determining the optimal intervals and their proportions. In order to validate our CPBIS mechanism in real environments, we evaluate it on the nRF52832 chip. The experimental results demonstrate that CPBIS can decrease the weighted average discovery latency 31.3%, and improve the success rate by 7.3%, compared with the baseline method ElastiCast-SBP. This validates the effectiveness of the CPBIS mechanism.

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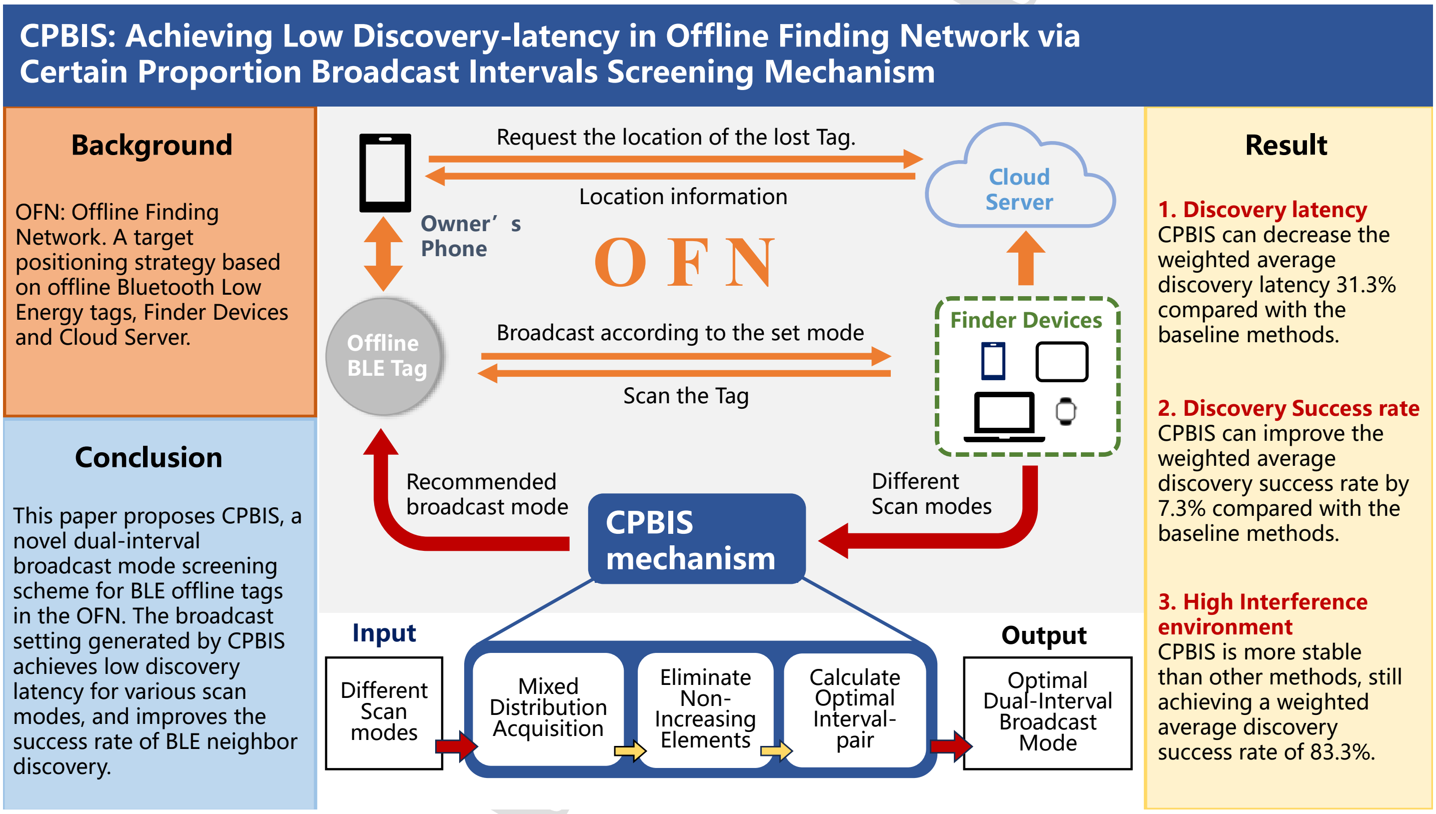
CPBIS: Achieving Low Discovery-latency in Offline Finding Network via Certain Proportion Broadcast Intervals Screening Mechanism
Background
OFN: Offline Finding Network. A target positioning strategy based on offline Bluetooth Low Energy tags, Finder Devices and Cloud Server.
Conclusion
This paper proposes CPBIS, a novel dual-interval broadcast mode screening scheme for BLE offline tags in the OFN. The broadcast setting generated by CPBIS achieves low discovery latency for various scan modes, and improves the success rate of BLE neighbor discovery.
Request the location of the lost Tag.
Location information
Cloud Server
Owner's Phone
OFN
Offline BLE Tag
Broadcast according to the set mode
Scan the Tag
Finder Devices
Recommended broadcast mode
Different Scan modes
CPBIS mechanism
Input
Different Scan modes
Mixed Distribution Acquisition
Eliminate Non-Increasing Elements
Calculate Optimal Interval-pair
Output
Optimal Dual-Interval Broadcast Mode
Result
1. Discovery latency
CPBIS can decrease the weighted average discovery latency 31.3% compared with the baseline methods.
2. Discovery Success rate
CPBIS can improve the weighted average discovery success rate by 7.3% compared with the baseline methods.
3. High Interference environment
CPBIS is more stable than other methods, still achieving a weighted average discovery success rate of 83.3%.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: